Orthogonal, Measured

A Preprint

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Abstract

The brief: two reservoirs, one "orthogonal" to the other — with the challenge of deciding what orthogonal should mean. We made it a number. Two half-width reservoirs (N=512 each) run in parallel on the identical text stream, states concatenated to width 1024; pond B is either a *bit-identical duplicate* of pond A or an *independent draw*; orthogonality is measured by canonical correlation between the two 512-dimensional blocks. Two findings, each hit on all three seeds, and one registered methodological observation. (1) A duplicate twin is a pure reparametrization, **exactly**: it ties the lone half-pond on ridge bpc and decode depth to the last recorded digit, three seeds in a row. (2) An independent twin beats the duplicate on **every** instrument — ridge, logistic, and decode depth — and lands within 0.0008–0.0049 bits of a full-width 1024-unit pond built as one piece: near-free doubling. (3) The observation: the naive orthogonality reading is a trap the measurement caught — the two ponds' *single best-aligned* directions correlate above 0.999 whether or not the wiring was copied, because shared input dominates the top of the spectrum; only the *full-spectrum* mean canonical correlation (≈0.90 duplicate vs ≈0.45 independent at full budget, recorded on two of the three seeds) distinguishes a copy from a genuinely new pond, and predicts which one pays downstream.

Keywords reservoir computing · canonical correlation analysis · redundancy · ensembles · feature width · pre-registration

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§1 The question

When does a second reservoir add information to a first one? Folk intuition says: when it is "orthogonal" — differently wired, independently random. But two reservoirs driven by the same input are not independent systems; both spend most of their capacity linearly tracking the same highly-recoverable recent-character history. The brief demanded a creative interpretation of orthogonality; the honest interpretation turned out to be an *operationalized* one — pick a measure, then check whether the measure predicts the downstream payoff of concatenating the two ponds' states.

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§2 The machine

Two house sparse-mixing tanh half-ponds ($N=512$ each, $\rho=0.95$, fanin 10, house input map, leak 1.0) [1] run in parallel on the identical text8 stream [3], 2M train characters; their states are concatenated to readout width 1024, matched to the standard full-width pond of the addressed-pond study so no feature-width confound enters. Three cells per seed: **single512**, one bare half-pond; **twin_redundant**, pond B built from pond A's own seed — identical wiring, bit-identical states every step, the maximally non-orthogonal case; **twin_orthogonal**, pond B an independent draw. Orthogonality is computed, not assumed: canonical correlation analysis [2] between the two 512-dim state blocks on subsampled validation rows ($\leq$100k, ridge-regularized whitening), summarized two ways — `top_canon_corr`, the single best-aligned direction, and `mean_all_canon_corr`, the average over all 512 canonical directions. The full-width comparator (ridge 3.1472, logistic 2.6727, depth 10.11) is the archived champion cell, cited, not rerun. Instruments: ridge and logistic validation bpc, U3 decode depth.

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§3 What we registered in advance

The pre-registration smoke held a genuine surprise, disclosed in the registration before the full run: *independent draws do not decorrelate the top of the spectrum.* The predictions were written with that disclosed:

- **O1/O2** (manipulation checks): the full-spectrum mean separates the twins by a wide margin, while the top canonical correlation stays ≥ 0.99 in *both* twin arms — with the named alternative that more data at full budget resolves the saturation away, in which case it is a smoke artifact. (Both hit; the saturation is real at 20× the data.)
- **O3** (reparametrization null): the duplicate twin ties the lone half-pond on ridge bpc ($|\text{gap}| \leq 0.01$) and decode depth ($|\text{gap}| \leq 0.3$). (Hit — exactly, gap 0.0000 on both.)
- **O4** (the headline): the independent twin beats the duplicate on all three of ridge, logistic, and depth. Named alternative: any single reversal = the smoke's pattern was small-sample noise. (Hit, 3/3.)
- **O5** (near-free doubling): the independent twin lands closer to the full-width comparator than the lone half-pond does, on both bpc instruments. (Hit.)

Quantitative bands Q1–Q4 were registered separately; two hit and two missed *in the beat-expectations direction* — the 100k-char smoke undersold how far the logistic readout converges at 2M characters, so both logistic bands' actuals came in below their registered floors. Recorded as misses, as the lab's "numbers usually miss" pattern predicts. The seed-1 rerun registered O3/O4/O5 unbundled (each promoting alone); a third seed was then requested through the lab dashboard and run under the identical registered wording.

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§4 What happened

Table 1. All three seeds (run ledger #18 / #19 / #25; values seed 0 / 1 / 2). Full-width comparator, cited: ridge 3.1472, logistic 2.6727, depth 10.11. Zero gate failures, zero AUTO-VOIDs.

cell	ridge val bpc	logistic val bpc	U3 depth
single512	3.3142 / 3.3100 / 3.3013	2.8356 / 2.8318 / 2.8266	8.66 / 8.57 / 8.18

twin_redundant	3.3142 / 3.3100 / 3.3013	2.8330 / 2.8289 / 2.8229	8.66 / 8.57 / 8.18
twin_orthogonal	**3.1442 / 3.1487 / 3.1423**	**2.6713 / 2.6735 / 2.6714**	**10.08 / 10.18 / 10.03**

The duplicate is inert, to the digit. On every seed, twin_redundant's ridge bpc and decode depth equal single512's exactly as recorded — 3.3142/8.66, 3.3100/8.57, 3.3013/8.18 — the cleanest null this lab produces. A duplicate readout column adds zero information a linear instrument can use; this is the 2×-redundancy edge case of the lab's measured widening-is-a-reparametrization fact, now with an exact copy instead of a permuted view.

The independent twin wins everything, three times. Ridge 3.1442/3.1487/3.1423 vs the duplicate's 3.3142/3.3100/3.3013; logistic 2.6713/2.6735/2.6714 vs 2.8330/2.8289/2.8229; depth 10.08/10.18/10.03 vs 8.66/8.57/8.18. And it sits 0.0030/0.0015/0.0049 ridge-bits and 0.0014/0.0008/0.0013 logistic-bits from the full-width comparator, where the lone half-pond sits 0.15–0.17 bits away on both — at least thirty times closer, on every seed and instrument. Two independently-drawn half-ponds recover almost everything one big pond gets.

The meter's top note saturates. top_canon_corr: 0.9999 (duplicate, every seed) vs 0.9995/0.9994/0.9995 (independent) — indistinguishable in practice. The full-spectrum mean separates them decisively: 0.9060/0.9022 vs 0.4477/0.4441 on the two seeds where the full-budget run recorded it (the third seed's verdict table recorded the top-direction summary only; its smoke showed the same shape, 0.8967 vs 0.4562). Registered as a methodology observation and carried in the lab's intake notes rather than the findings book: it is a point about how to *measure* orthogonality between co-driven reservoirs, not an effect size.

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§5 What it means

Redundancy is exactly worthless; independence is nearly free. If you want a second reservoir to add value, drawing it independently buys almost the whole doubling: two half-ponds recover a full pond to within thousandths of a bit, three seeds in a row. Copying the wiring buys exactly nothing, to the decimal, three seeds in a row. The result composes with the goat-herd finding from the same day's program — there, 64 blocks of one draw at matched width never cost and usually helped; here, two fully independent draws (own mixing matrix *and* own input map) nearly match a monolithic pond.

"Orthogonal" is a measurement, not a construction. Two ponds with independent wiring *sound* orthogonal, but their best-aligned directions correlate above 0.999, because the shared input dominates what both ponds most strongly represent. Independence only shows up — and only pays — across the rest of the spectrum. Any pipeline that certifies diversity by construction ("different seeds, therefore decorrelated") is measuring the wrong end of the canonical spectrum.

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§6 References

1. H. Jaeger, "The 'echo state' approach to analysing and training recurrent neural networks," GMD Report 148, German National Research Center for Information Technology, 2001.
2. H. Hotelling, "Relations between two sets of variates," *Biometrika* 28, 1936.
3. M. Mahoney, "Large text compression benchmark" (text8), mattmahoney.net/dc/textdata.

§7 Provenance

- **c660a423** — `2026-08-11-caity-orthogonal-reservoir`: the brief; 3 cells, seed 0 (run ledger #18). Registered before running; the smoke's saturation surprise disclosed in the registration. Headline hit 3/3 ⇒ provisional, replication auto-queued before any findings entry.
- **142d0104** — `2026-08-11-orthogonal-reservoir-rerun1`: seed 1, fresh draws of both ponds (run ledger #19). All three unbundled claims hit; promoted.
- **8879faae** — `2026-08-11-caity-orthogonal-reservoir-rerun`: seed 2, requested through the lab dashboard as a direct third-seed confirmation; identical registered wording (run ledger #25). All three claims hit again; findings annotated 3/3 seeds.

All claims judged strictly against the registered wording, misses included above; 3 seeds; replicated before publication.